

Daniel Campos Zamora

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Education

University of Washington

2021-Present

PhD, Computer Science and Engineering

Advisor: Jon Froehlich

University of Washington

2021-2024

M.S. in Computer Science and Engineering

Carnegie Mellon University

2011-2015

BHA in Cognitive Psychology and Media Art

Professional Experience

Adobe Research

Summer 2023, 2024

Research Scientist Intern

Researched passive tangible interaction using computer vision (2023) and haptics (2024) with 3D printed optical and vibratory markers to detect and decode user inputs.

Advisors: Dr. Chang Xiao, Dr. Doga Dogan, Dr. Eunye Koh

Disney Research / Walt Disney Imagineering

2017-2019

Research Associate

Developed soft, tangible interfaces for physical human-robot interactions, haptics for social VR/AR experiences, and lifelike robot gaze behaviors.

Supervisors: Dr. Katsu Yamane, Dr. Joohyung Kim

Carnegie Mellon Social and Decision Sciences

Summer 2016

Research Associate

Prototyped wearable acoustic recording device to study the link between the gut and mind with novel signal processing techniques.

Principal Investigators: Dr. George Loewenstein, Dr. Ali Momeni

Publications

Refereed Publications

- [C-9] **Daniel Campos Zamora**, Ria Agarwal*, Raiden Santos*, Jerry Cao, Jon E. Froehlich, Chang Xiao. 2026. FlexiCodes: Embedded De Bruijn Patterns for Deformation Tracking on Elastic 3D-Printed TPU Objects with a Monocular Camera (*Under Review*)
- [C-8] **Daniel Campos Zamora**, Richard Li, Doga Dogan, Joseph Breda, Eunye Koh, Jon E. Froehlich, Chang Xiao. 2026. HapticWidgets: Passive Haptics as an Input/Output Mechanism for Tangible Interaction via Wearable Sensing. (*Under Review*)
- [C-7] Kenta Hirashima, **Daniel Campos Zamora**, Kevin Gim, Joohyung Kim. 2026. "Self-rising Bipedal Robot for Embracing Fall Impact and Fall Detection with Multimodal Sensing," 2026

IEEE/SICE International Symposium on System Integration (SII), Cancun, Mexico, 2026, pp. 339-344. <https://doi.org/10.1109/SII64115.2026.11404570>

- [C-6] Joseph Breda*, **Daniel Campos Zamora***, Shwetak Patel, and Jon E. Froehlich. 2025. NightLight: Passively Mapping Night-time Sidewalk Light Data for Improved Pedestrian Routing. In Proceedings of the 43rd Annual ACM Conference on Human Factor in Computing Systems (CHI '25). Yokohama, Japan 2025. <https://doi.org/10.1145/3706598.3714299>
- [C-5] **Daniel Campos Zamora**, Doga Dogan, Alexa Siu, Eunyee Koh, Chang Xiao.2024. MoiréWidgets: High-Precision, Passive Tangible Interfaces via Moiré Effect. In Proceedings of the 42nd Annual ACM Conference on Human Factor in Computing Systems (CHI '24). Honolulu, Hawaii, USA 2024. <https://doi.org/10.1145/3613904.3642734>
- [C-4] Kenta Hirashima, Noboru Myers, **Daniel Campos Zamora**, Kevin Gim, Joohyung Kim. 2024. "Implementation of Untethered Biped Robots Utilizing Serial-Parallel HybridLeg Mechanisms," 2024 IEEE-RAS 23rd International Conference on Humanoid Robots (Humanoids), Nancy, France, 2024, pp. 227-233. <https://doi.org/10.1109/Humanoids58906.2024.10769967>
- [C-3] **Daniel Campos Zamora**, Liang He, and Jon E. Froehlich. 2024. MobiPrint: A Mobile 3D Printer for Environment-Scale Design and Fabrication. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST '24). Association for Computing Machinery, New York, NY, USA, Article 38, 1–10. <https://doi.org/10.1145/3654777.3676459>
- [C-2] Matthew K.X.J. Pan, Sungjoon Choi, James Kennedy, Kyna McIntosh, **Daniel Campos Zamora**, Gunter Niemeyer, Joohyung Kim, Alexis Wieland, and David Christensen. 2020. Realistic and Interactive Robot Gaze. In 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, Las Vegas, NV, USA, 11072–11078. <https://doi.org/10.1109/IROS45743.2020.9341297>
- [C-1] Eric Sebastian Sanchez, **Daniel Campos Zamora**, and Joohyung Kim. 2019. Development of Airbag Fabrication Machine and Process for Physical Human Machine Interaction. In 2019 2nd IEEE International Conference on Soft Robotics (RoboSoft), IEEE, Seoul, Korea (South), 502–508. <https://doi.org/10.1109/ROBOSOFT.2019.8722726>

Symposiums and Workshop Organizing

- [DS-1] *Augmenting Physical World Interactions across Material Scale from Artifacts to Environments*. 39th Annual ACM Symposium on User Interface Software and Technology ('26 UIST Adjunct Proceedings).
- [W-2] *Health Centric Wearable Devices in the Age of Physical AI*. UbiComp '26: The 2026 ACM International Joint Conference on Pervasive and Ubiquitous Computing / ISWC ACM International Symposium on Wearable Computers.

Poster and Workshop Papers

- [P-4] **Daniel Campos Zamora**, Liang He, and Jon E. Froehlich. 2024. Towards Rapid Fabrication of Custom Tactile Surface Indicators for Indoor Navigation. In Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24). Association for Computing Machinery, New York, NY, USA, Article 97, 1–5. <https://dl.acm.org/doi/10.1145/3663548.3688519>
- [P-3] Xia Su, **Daniel Campos Zamora**, and Jon E. Froehlich. 2024. RAIS: Towards A Robotic Mapping and Assessment Tool for Indoor Accessibility Using Commodity Hardware. In

Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24). Association for Computing Machinery, New York, NY, USA, Article 86, 1–5 <https://doi.org/10.1145/3663548.3688512>

- [P-2] Mia Hoffman, **Daniel Campos Zamora**, Jon E. Froehlich, Katherine M. Steele, and Heather A. Fender. “Turn right here”: A Quantitative Approach to Testing the Efficacy of Novel Steering Modifications for Adapted Ride-on Cars. 38th International Seating Symposium (ISS). Pittsburgh, Pennsylvania, USA 2023 <https://doi.org/10.1080/10400435.2023.2198437>
- [W-1] Aashaka Desai, Venkatesh Potluri, Aaleyah Lewis*, **Daniel Campos Zamora***, Jayne Everson*, Jennifer Mankoff, and Richard E. Ladner, “Using Fiber Arts and Sonification to Improve Data Accessibility of Maker Spaces,” In Proceedings of the 40th Annual ACM Conference on Human Factor in Computing Systems (CHI '22). New Orleans, LA USA.
- [P-1] **Daniel Campos Zamora**, Liang He, Yueqian Zhang, Xuhai Xu, Jennifer Mankoff, and Jon E. Froehlich. 2022. sPrintr: Towards In-Situ Personal Fabrication using a Mobile 3D Printer. In Symposium on Computational Fabrication, ACM, Seattle WA USA, 1–3. <https://doi.org/10.1145/3559400.3565587>

Invited Talks

<i>Augmented Tangible and Embodied Interaction</i> UCLA HCI	October 2026
<i>Diseño y fabricación para la interacción tangible con objetos y entornos físicos</i> Universidad de Costa Rica	June 2026
<i>Design and Fabrication for Tangible Interaction with Real-World Objects and Environments</i> University of Maryland	May 2025
<i>MobiPrint and MoiréWidgets</i> Boston University Graphics Lab	November 2024
<i>MobiPrint</i> UW CSE Accessibility Course	October 2024

Patents

“Eye contact sensing and control for robotic characters.” U.S. Patent US11072075

Service and Leadership Experience

<i>Reviewer</i> ACM CHI	2024, 2025, 2026
<i>Student Coordinator</i> DUB @ University of Washington	2023-Present
<i>Graduate Student Leader</i> NASA SURP Program UW	2024
<i>Graduate Representative</i> UW CSE Diversity and Inclusion Committee	2022-2023

Graduate Outreach and Undergrad Liaison 2022-2023
UW Computer Science and Engineering

Student Volunteer 2022
ACM CHI

Volunteer Translator 2018-2021
Esperanza Immigrant Rights Project

Chairman 2014-2015
Spirit Racing Systems

Honors & Awards

Graduate Student Fellowship 2026-2027
NASA Washington Space Grant

SEED Grant (\$40,000) 2024
UW CREATE

1st Place Graduate Student Research Competition 2024
ACM TAPIA

Graduate School Conference Presentation Award 2024
UW Graduate School Office

Travel Grant 2024
UW Graduate and Professional Student Senate

Travel Grant 2024
UW Allen School of Computer Science and Engineering

GEM Fellow 2023
National GEM Consortium

Graduate Cohort for IDEALS Travel Grant 2023
UW Allen School of Computer Science and Engineering

Rising Doctoral Institute Candidate 2023
Rising Doctoral Institute

Senior Leadership Recognition Award 2015
Carnegie Mellon University

Summer Undergraduate Fellowship 2013
Carnegie Mellon University

Teaching Experience

Teaching Assistant 2022, 2025, 2026
MHCI 521: Prototyping Studio

Teaching Assistant
CSE 340: Interaction Programming

2022

Teaching Assistant
INFO 300: Research Methods

2021

Selected Press

Tech Radar	Someone fused a 3D printer and a vacuum robot to create a mobile 3D printer that can print on almost any flat surface — including your floor
IEEE Spectrum	This mobile 3D printer can print directly on your floor
GeekWire	3D printing on the move: UW device can map a room and print custom items in desired space
TechXplore	Meet MobiPrint: A smart 3D printer that can upgrade your home instantly
3Printr	MobiPrint: Researchers present mobile 3D printer
Hackster.io	MobiPrint hits the ground running
IEEE Spectrum	Disney research makes robotic gaze interaction eerily lifelike
Business Insider	Disney unveiled an animatronic robot with the same gaze as humans built to create 'the illusion of life'
Popular Mechanics	Watch this Disney robot make the most convincing eye contact ever
CNET	Disney figured out how to make a robot gaze into your soul
Engadget	Disney robot with human-like gaze is equal parts uncanny and horrifying
CNN	Disney's new robot has a creepily human gaze
DesignBoom	Disney research unveils a skinless humanoid robot with creepy lifelike eyes
Gizmodo	Disney made a skinless robot that can realistically stare directly into your soul
The Verge	Disney's new skinless robot can blink like a human because why not

Skills

Programming

Python, C++, Arduino, Java, Processing, ROS, Unity/C#

Research Methods

Experiment Design, User Testing, A/B testing, Experience Prototyping

Fabrication

CAD/CAM, 3D Printing, Laser Cutting, Woodworking, Sewing

Design

Solidworks, Rhino3D, Fusion360, KeyShot, RhinoCAM, Unity3D, SketchUp, Adobe CS